

**Sixth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

OBJECT ORIENTED DATA STRUCTURE

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Assume suitable data wherever required and mention it.

1. (a) (1) What is wrong with each of the following commands ?
Javac Example.class
Java Example.class
(2) Does it matter where on a line you put a statement in java ?
2(CO1)
- (b) What do you mean by Typecasting in java ? Explain types of it with example.
4(CO1)
- (c) The sum of first n natural numbers is given by : $\text{sum} = n(n + 1) / 2$. Write a java program to print the sum of first n natural numbers.
4(CO1)
2. (a) Write a java program for binary search, numbers to be entered by command line.
4(CO2)
- (b) Create a **Person** class with private instance variable for the person's name and birth date. Add appropriate accessor methods for these variables. Then create a subclass **CollegeGraduate** with private instance variable for the students' GPA and year of graduation and appropriate assessors for these variables. Add appropriate constructors for your classes. Then create a class **System** with a main method that demonstrate your classes. 4(CO2)
- (c) Is there any difference between parameterized constructor and overloaded constructor ?
2(CO2)

3. (a) What do you mean by compile time polymorphism ? How it is different than run time polymorphism ? 5(CO2)
- (b) Create a user defined package name homebudget which contains home expenditure and Home income class. For both the classes find a suitable data member and its functionality. Define another class where you have to import home budget package and find out the total savings of your family. 5(CO2)

4. (a) Prove that running Time (any **Three**) :—

(i) $T(n) = n^3 + 50n$ is $\Omega(n^2)$

(ii) $T(n) = n^3 + 25n + 5$ is $O(n^4)$

(iii) $T(n) = n^3 + 20n + 3$ is not $O(n^2)$

(iv) $T(n) = n^4 + 20n^2 + 6$ is $\Omega(n^3)$

Using suitable examples.

6(CO3)

- (b) Find the complexity for the given code and elaborate.

(i) $k=0$ For ($i = 0 ; k < n ; i++$) { $k = k + i ;$ }	(ii) $\text{for } (i = n ; i > 0 ; i--)$ { $\text{for } (j = 1 ; j < n ; j * 2)$ { $\text{cout} << i ;$ } }
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4(CO3)

5. (a) Write a program in JAVA to implement circular queue. Interface should have three methods for inserting, Deleting and displaying the items in circular queue.

OR

Write a program in JAVA to implement stacks operation. Interface should have methods for Push and Pop operation. 5(CO4)

- (b) What is the value of the postfix expression 6 3 2 4 * ? 11 40 ?
Evaluate the above expression using stack operation. Also write an algorithm for the same. (Infix to postfix) 5(CO4)
6. (a) Write an algorithm to find particular element from a singly linked list. What will be the time complexity of this operation ? 5(CO4)
- (b) Which are the advantages of Doubly linked list over simple linked list ?
Write a java method to add an element in the middle of doubly linked list. 5(CO4)

